Java MCQ Assignment – 20 Questions

Topics: Introduction to Java, Data Types, Variables, Operators, Conditional & Control Statements, Methods, Arrays

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1. Which statement best explains Java's platform independence?

A. Java programs are directly converted into machine code
B. Java source code is interpreted by the operating system
C. Java bytecode can run on any system having a compatible JVM
D. Java does not require a compiler

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2. Which component is responsible for executing Java bytecode?

A. JDK
B. JRE
C. JVM
D. JAR

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3. Which of the following is NOT a valid Java identifier?

A. `studentName`
B. `_studentName`
C. `$studentName`
D. `student-name`

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4. What is the output of the following code?

```
int a = 10;
int b = 3;

System.out.println(a / b);
System.out.println(a % b);
```

A. `3.33` and `1`
B. `3` and `1`
C. `3` and `0`
D. `4` and `1`

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5. What is the output?

```
int x = 5;

System.out.println(x++);
System.out.println(++x);
System.out.println(x);
```

A. `5 6 6`
B. `5 7 7`
C. `6 7 7`
D. `5 6 7`

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6. Which conversion is valid without explicit type casting?

A. `double` → `int`
B. `long` → `short`
C. `int` → `long`
D. `float` → `byte`

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7. What will be the output of the following code?

```
int a = 5;
int b = 10;

boolean result = (a++ > 5) || (++b > 10);

System.out.println(a);
System.out.println(b);
System.out.println(result);
```

A. `6 11 true`
B. `6 10 true`
C. `5 11 false`
D. `5 10 false`

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8. What is the output of the following program?

```
int x = 15;

if (x > 10)
    if (x > 20)
        System.out.println("A");
    else
        System.out.println("B");
else
    System.out.println("C");
```

A. A
B. B
C. C
D. Compilation error

---

9. What is the output?

```
int x = 10;

if (x > 5 && x++ > 10)
    System.out.println("Yes");
else
    System.out.println("No");

System.out.println(x);
```

A. `Yes 11`
B. `No 11`
C. `No 10`
D. `Yes 10`

---

10. What will be the output of the following Java code?

```
int n = 5;

while(n > 0) {
    System.out.print(n + " ");
    n = n - 2;
}
```

A. `5 4 3 2 1`
B. `5 3 1`
C. `4 2`
D. `5 3 2 1`

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11. What is the output?

```
for(int i = 1; i <= 5; i++) {

    if(i == 3)
        continue;

    System.out.print(i + " ");
}
```

A. `1 2 3 4 5`
B. `1 2 4 5`

C. `1 2`
D. `3 4 5`

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12. Which of the following statements about a Java method is correct?

A. A method must always return a value
B. A method can return only primitive data types
C. A method can accept parameters and may or may not return a value
D. A method cannot be called more than once

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13. Consider the following method:

```
static int calculate(int a, int b) {
    return a + b;
}
```

Which statement correctly calls the method and stores the result?

A. `calculate = (10, 20);`
B. `int result = calculate(10, 20);`
C. `int result = calculate;`
D. `calculate(10, 20) = result;`

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14. What will be printed?

```
static void change(int x) {
    x = 100;
}

public static void main(String[] args) {

    int a = 10;

    change(a);

    System.out.println(a);
}
```

A. 10
B. 100
C. 0
D. Compilation error

---

15. What is the output?

```
static int test(int x) {

    if(x > 10)
        return x * 2;

    return x + 2;
}

public static void main(String[] args) {

    System.out.println(test(5));
    System.out.println(test(15));
}
```

A. `7 17`
B. `10 30`
C. `7 30`
D. `10 17`

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16. What is the output?

```
int[] a = {10, 20, 30, 40, 50};

System.out.println(a.length);
System.out.println(a[2]);
```

A. `4` and `30`
B. `5` and `20`
C. `5` and `30`
D. `6` and `30`

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17. What happens when the following code is executed?

```
int[] a = new int[5];

a[5] = 100;
```

A. 100 is stored successfully
B. Compilation error
C. `ArrayIndexOutOfBoundsException` occurs
D. `NullPointerException` occurs

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18. What is the output?

```
int[] a = {2, 4, 6, 8};

for(int i = 0; i < a.length; i++) {
```

```
    if(a[i] % 4 == 0)
        a[i] = a[i] + 1;
}

System.out.println(a[1] + " " + a[3]);
```

A. `4 8`
B. `5 9`
C. `4 9`
D. `5 8`

---

19. What will be the output?

```
static void modify(int[] a) {

    a[0] = 100;
}

public static void main(String[] args) {

    int[] arr = {10, 20, 30};

    modify(arr);

    System.out.println(arr[0]);
}
```

A. 10
B. 20
C. 30
D. 100

---

20. What is the output?

```
static int calculate(int[] a) {

    int sum = 0;

    for(int i = 0; i < a.length; i++) {

        if(a[i] % 2 == 0)
            sum += a[i];
    }

    return sum;
}

public static void main(String[] args) {

    int[] a = {5, 10, 15, 20, 25, 30};

    System.out.println(calculate(a));
```

```
}
```

A. 45
B. 50
C. 60
D. 75
